

Smart Compaction: Predicting Compaction Utility from Lakehouse Table Metadata

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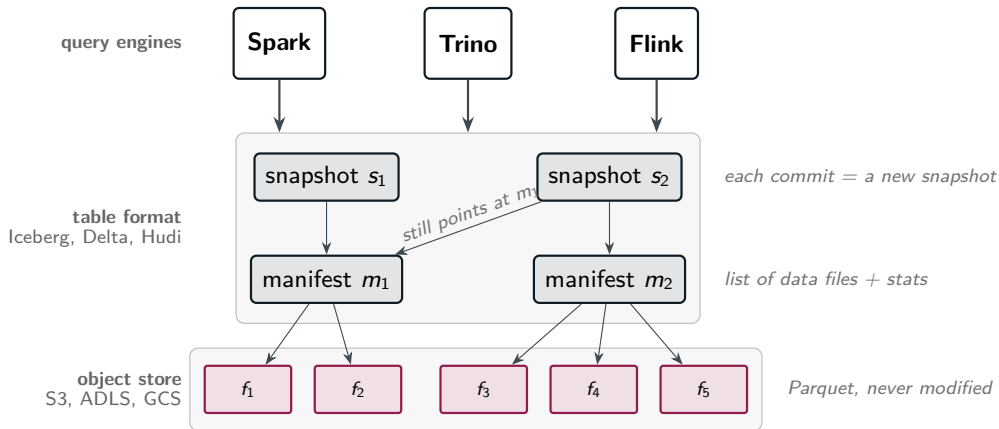
Smart Data Processing track

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Code: <https://github.com/JannicCutura/smart-compaction>

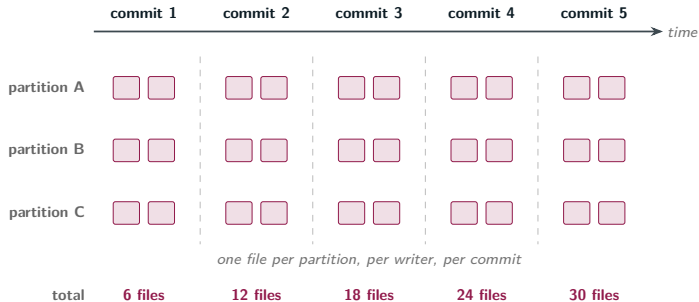
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A lakehouse table is a set of immutable files plus a metadata tree



Engines see a table through its metadata, not its files. Every commit **appends** new files and a new snapshot and never touches existing ones — that is what buys ACID, time travel and snapshot isolation on cheap object storage.

Every commit appends files, nothing ever merges them



files = partitions × writers × commits.

Our grid tops out at $100 \times 5 \times 20 = 10,000$ files for a single table
— most of them far below the size a scan wants.

Why file count hurts



Planning

every manifest entry is read per query



Object store

LIST and GET are billed per request

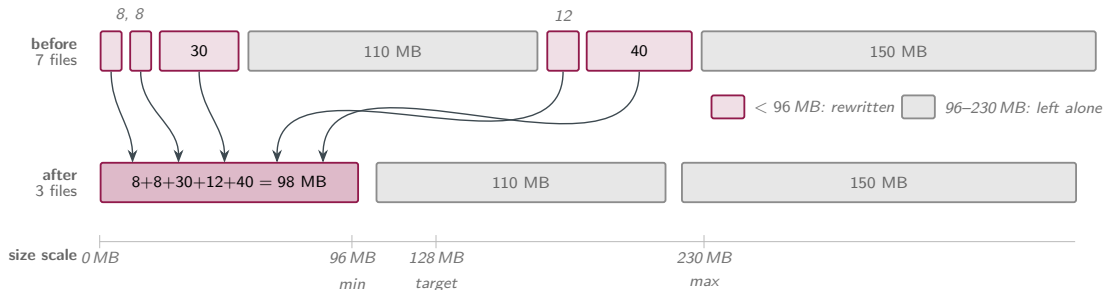


Scan

open + footer read + task, per file

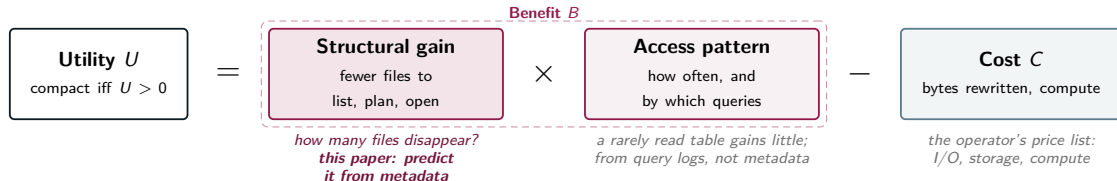
At scale: LinkedIn's AutoComp manages 35,000+ Iceberg tables; catalogs accumulate **millions** of undersized files.

Compaction rewrites many small files into a few target-sized ones



Iceberg's `rewrite_data_files` with **bin-packing**: files below the minimum are packed into target-sized outputs; files already in range are skipped. Reading and rewriting those bytes is the price — so the question is not *whether* to compact, but *when it pays*.

Compact or not? Utility is benefit minus cost



We measure the structural gain as the **file-reduction ratio**

$$r_{\text{fc}} = 1 - \frac{n_{\text{after}}}{n_{\text{before}}} \in [0, 1]$$

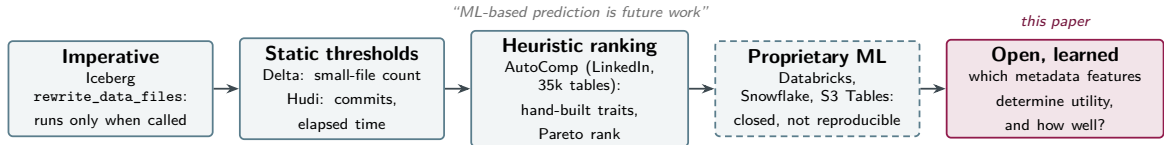
$r_{\text{fc}} \approx 1$: the table collapses to a few files. $r_{\text{fc}} \approx 0$: the rewrite is wasted I/O. On the previous slide, $r_{\text{fc}} = 1 - 3/7 = 0.57$.

Two questions for the metadata

- 1 **Whether:** will compaction rewrite anything at all?
- 2 **How much:** what r_{fc} will it achieve?

Access pattern and cost are the operator's; we supply the structural signal.

Today the decision is a rule; we make it a prediction



Every open scheduler fires on a fixed rule; no published study asks *which* metadata determines compaction utility.

Target

$$r_{fc} = 1 - \frac{n_{\text{after}}}{n_{\text{before}}} \in [0, 1]$$

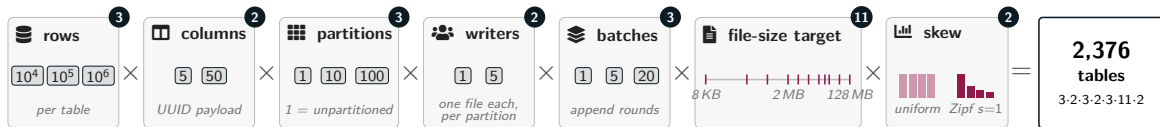
measured by running Iceberg's bin-packing
rewrite_data_files on every table

Two tasks, from metadata only

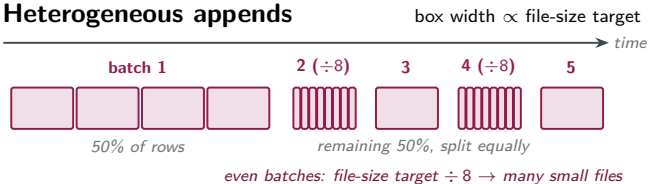
- 1 **Classification:** will bin-packing rewrite any file?
- 2 **Regression:** predict r_{fc} , for cost-aware scheduling

Inputs: manifests, snapshots, partition stats; no data file is read.

Simulation: 2,376 synthetic tables, each compacted once and labelled

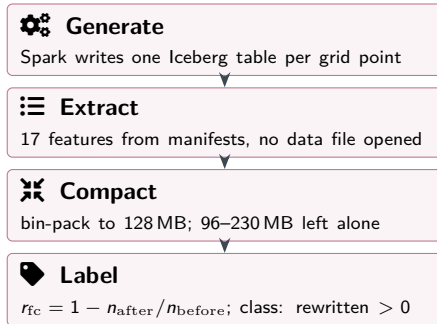


Heterogeneous appends

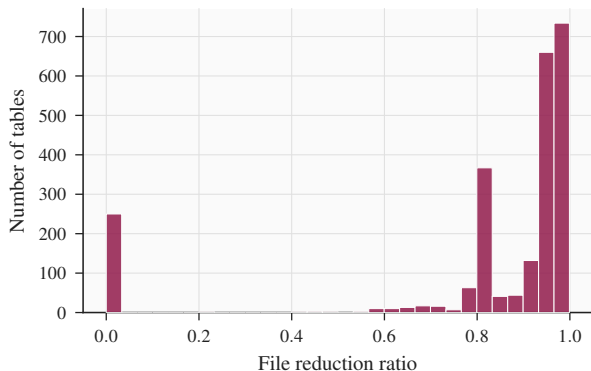


One table, two file-size regimes: mean file size alone no longer gives the label away — the **tautology trap**.

Per table



What the Simulation Produced



Distribution of r_{fc} across all 2,376 tables.

 **2,376** tables generated

 **2,121** (89.3%) had ≥ 1 file rewritten

 r_{fc} : mean **0.818**, median 0.950, $\sigma=0.294$

Bimodal, not continuous

Either compaction does essentially nothing, or it does a lot. Only **11 of 2,376** tables land in $(0, 0.5]$.

So the binary label and the continuous target nearly coincide — which is the first hint that *whether* is the easy question and *how much* is the interesting one.

Classification: Binary Decision

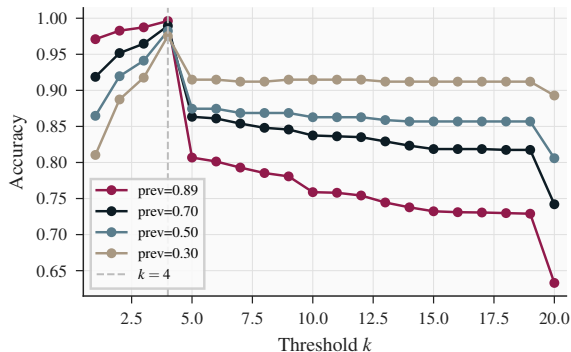
Classification (held-out test, $n = 476$)

Model	Acc.	F1
XGBoost	1.000	1.000
Logistic Reg.	0.815	0.885
Threshold ($k=4$)	1.000	1.000

The whole classifier

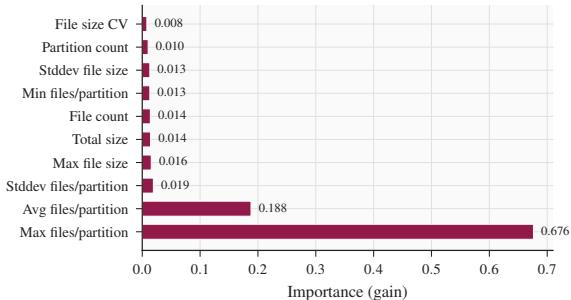
```
max_files_per_partition > 4
```

A single threshold matches XGBoost exactly. No learned model required.

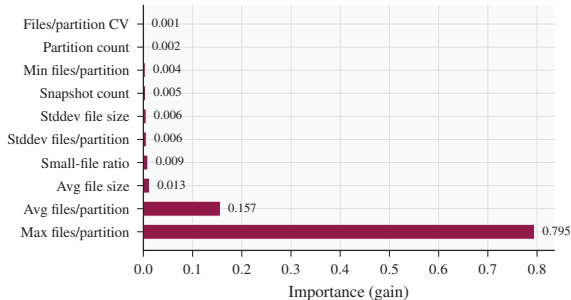


Accuracy peaks at $k = 4$ and collapses at $k = 5$ — a hard boundary, not a fitted one. Holds at every prevalence level.

Where the Signal Lives



Classifier — **86.4%** of gain



Regressor — **95.2%** of gain

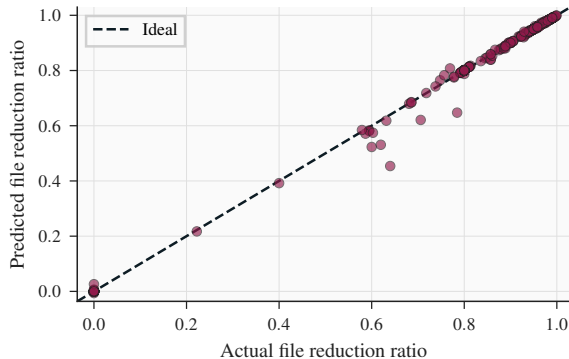
Two partition-level counts dominate both tasks. Every other feature contributes ≤ 0.019 gain.

Regression: Continuous Prediction

Regression (held-out test)

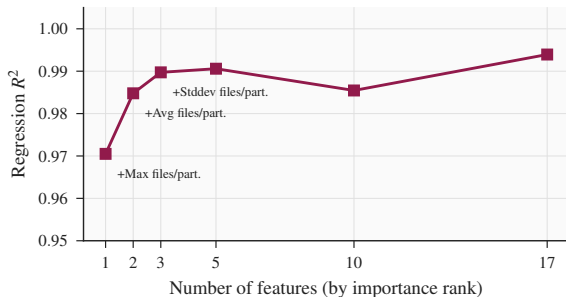
Model	RMSE	R^2
XGBoost	0.013	0.998
OLS	0.236	0.361

- OLS cannot capture the nonlinear saturation curve
- Two features account for **95.2%** of regressor gain
- Spearman $\rho = 0.965$: monotonic but not linear



Predicted vs. actual r_{fc} on the held-out test set.

Is the Result Fragile?



One feature alone reaches $R^2 = 0.970$; two reach 0.985. Past five, returns diminish.

Four robustness checks

✂ Ablation

other 15 features add only $\Delta R^2 = 0.023$ (95% CI [0.004, 0.044])

⊖ Leave-one-out

every single removal keeps $R^2 \geq 0.988$

🔄 30 bootstrap splits

$R^2 = 0.999$, CI [0.996, 1.000]

⚖ Prevalence 89%→30%

regression $R^2 > 0.995$ throughout

Holds across model families too — random forest 0.998, decision tree 0.997. Any tree-based model suffices; only the linear baseline fails.

TPC-H Validation (No Retraining)

96 Iceberg tables from 6 TPC-H schemas (lineitem, orders, customer, part, partsupp, supplier)

- Realistic data types (strings, dates, decimals)
- 7–16 columns per schema
- Scale factor ~ 10 (468.8 M rows, 15 GB)

Cross-schema results

Task	Acc./RMSE	F1/ R^2
Clf (XGBoost)	0.979	0.987
Clf (Threshold $k=4$)	0.979	0.987
Clf ($sfr > 0$)	0.833	0.908
Clf (Majority class)	0.823	0.903
Reg (XGBoost)	0.058	0.976

Key findings:

- Transfers with **no retraining** — no TPC-H table was seen during training
- XGBoost and the $k=4$ threshold tie at **97.9%**, missing the same two borderline tables (3 files at ~ 50 MB)
- The learned model earns its keep on the *continuous* target ($R^2 = 0.976$), not the binary decision

Weakest schemas are `customer` and `partsupp` ($R^2 \approx 0.87$), which produce tables sitting closest to the compaction threshold.

Conclusion

- ① **Binary compaction decision** is trivially solvable by a single partition-level threshold (`max_files_per_partition > 4`)
- ② **Continuous reduction ratio** requires nonlinear modelling (XGBoost $R^2 = 0.998$ vs. OLS $R^2 = 0.361$)
- ③ **Transfers across schemas** without retraining: **97.9%** on TPC-H (threshold and XGBoost alike), regression $R^2 = 0.976$
- ④ Compaction utility is determined by **structural metadata**, not data content
- ⑤ Downstream benefit is **workload-dependent**: metadata-heavy queries speed up, full-scan aggregations slow down
- ⑥ **Practical rule**: compact when predicted $r_{fc} > 0.3$ — drops the lowest-utility jobs while keeping **99.8%** of the beneficial ones



Caveats

Validated on synthetic grids and TPC-H, not production clusters; one Copy-on-Write bin-packing target ($k=4$ tied to our 128/96 MB config). And a methodological warning from our own first pass: a benchmark built only from undersized synthetic files compacts *everything* completely, producing misleadingly perfect metrics — realistic file-size heterogeneity is essential to a fair test.

Broaden scope

-  Extend to Delta Lake and Hudi
-  Merge-on-read: delete-file compaction, beyond Copy-on-Write
-  Sort and z-order compaction strategies
-  Generalise beyond one compaction-target configuration

Toward production

-  Add query-workload features and cloud-cost metrics
-  Model temporal evolution — utility drifts as tables change between writes
-  Multi-table scheduling via contextual bandits
-  Validation on production Iceberg catalogs, at distributed-cluster scale

All code and data publicly available.

Thank You

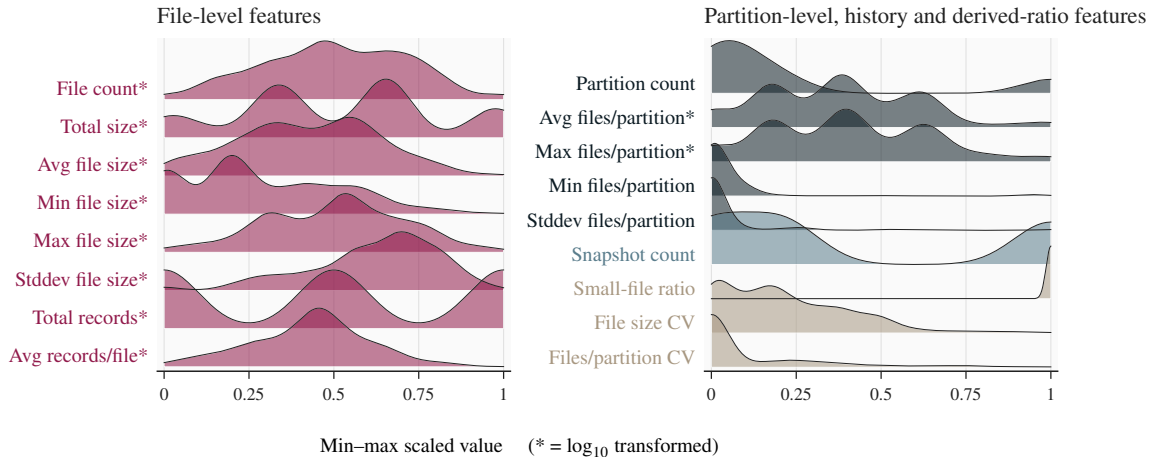
Questions?

Contact: `jannic.cutura@dsti.institute`

Simulation framework, models and code:

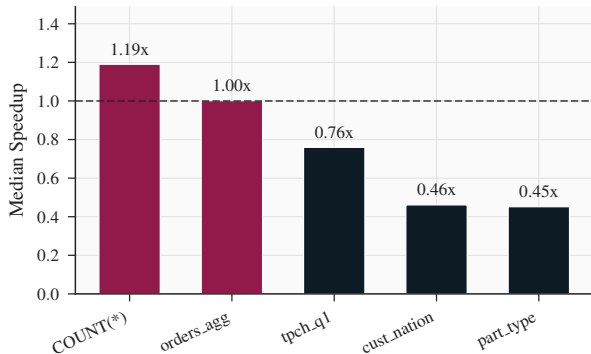
`https://github.com/JannicCutura/smart-compaction`

Backup: 17 Metadata Features



Min-max scaled to $[0, 1]$; * = $\log_{10}$ -transformed. Read from manifests alone — no data file is opened.

Backup: Does Compaction Actually Make Queries Faster?



52 TPC-H tables, Iceberg time-travel (VERSION AS OF) to query both snapshots. Median of 3 timed runs.

Not universally.

- COUNT(*): **1.19x** faster — metadata overhead dominates
- Filtered aggregation: neutral to 0.76x
- Full-scan aggregation: **0.45x** — 2x *slower*

? Why

Spark assigns tasks per file. Compacting 300 files into 1 drops parallelism from 300 tasks to 1, offsetting the metadata saving for queries that must read everything.

Overall median across 104 measurements: **0.97x**.

Backup: Per-Schema TPC-H Breakdown

Schema	n	Clf Acc.	Reg R^2
lineitem	16	1.000	0.997
orders	16	1.000	1.000
part	16	1.000	1.000
supplier	16	1.000	1.000
customer	16	0.938	0.874
partsupp	16	0.938	0.867

`customer` and `partsupp` each contribute one misclassified table. Both sit near the 96 MB compaction threshold, where a small file-size shift flips the outcome.

Backup: Query Latency by Type

Query	n	Median	Mean	ρ	p
count_star	52	1.19×	1.31×	0.50	<0.001
orders_agg	12	1.00×	1.63×	0.89	<0.001
tpch_q1	13	0.76×	0.82×	0.77	0.002
cust_nation	15	0.46×	0.56×	0.45	0.095
part_type	12	0.45×	0.47×	0.38	0.226

52 TPC-H tables, median of 3 timed runs after 1 warmup. Speedup $> 1\times$ means faster after compaction. ρ is the Spearman correlation with the file-reduction ratio.